Kestrel 3.0 release notes

Released 30 September 2026 · Download · Full changelog · Upgrade guide

Highlights

Kestrel 3.0 is the first major release in two years. It replaces the scenario engine, halves tail latency under load, and makes every run reproducible from a single `kestrel.lock` file.

47% lower p99 latency in the reference scenario, from 356 ms to 188 ms.

Reproducible runs: seeds, versions, and targets are pinned in `kestrel.lock`.

Scenario files in TOML, checked by `kestrel check` before a run starts.

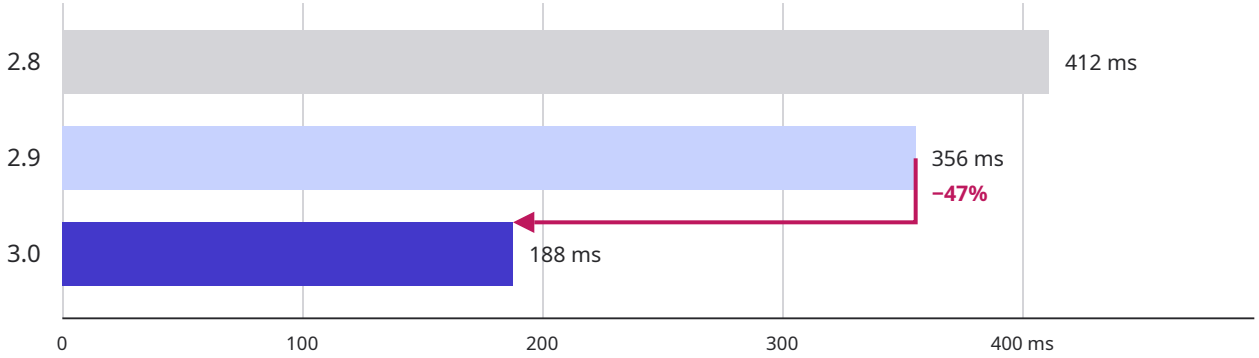

Figure 1. p99 latency in the reference scenario for 2.8, 2.9, and 3.0 (shorter is better)

3.0.0 · 30 September 2026

Breaking changes

Action required. YAML scenarios no longer load. Convert them before upgrading:

```
kestrel migrate scenarios/ --to toml --write
```

- Scenario files are TOML; `kestrel migrate` converts YAML files in place. (#2210)
- The `--rps` flag is now `--rate` and accepts units such as `500/s`. (#2187)
- Metrics are exported as OpenTelemetry by default; set `export = "statsd"` to keep the old format. (#2143)

- macOS on Intel is no longer supported; the 2.9 LTS line remains available. (#2231)

Added

- `kestrel.lock` pins the random seed, target versions, and plugin hashes for every run. (#2102)
- `kestrel check` validates scenarios, including unreachable steps and unused variables. (#2125)
- New open-loop arrival models:
  - `poisson` for independent arrivals; (#2166)
  - `ramp` and `step` for capacity searches. (#2167)
- A static musl build for minimal container images. (#2198)
- Live terminal dashboard, enabled with `--watch`. (#2204)

Fixed

- Coordinated omission no longer hides stalls longer than the sampling interval. (#2091)
- HTTP/2 connections are reused across scenario steps instead of per step. (#2118)
- Reports keep their percentile order when a run is interrupted with `Ctrl-C`. (#2150)

Deprecated

- The `--duration` flag; set `duration` in the scenario file instead. Removal is planned for 4.0. (#2215)
- The Graphite exporter, in favour of OpenTelemetry. (#2144)

Known issues

- The live dashboard flickers in terminals narrower than 80 columns. (#2240)
- `kestrel migrate` drops YAML comments; review converted files before committing them. (#2236)

Compatibility

Kestrel 3.0 raises the minimum versions on most platforms. Check Table 1 before you upgrade agents on older hosts.

Table 1. Supported platforms and minimum versions

Platform	2.9	3.0	Notes
Linux x86-64 (glibc)	2.28	**2.31**	Ubuntu 20.04, RHEL 9 and later
Linux arm64 (glibc)	2.28	**2.31**	Graviton and Ampere tested
Linux x86-64 (musl)		**1.2**	New static build for containers
macOS arm64	12	**13**	Ventura and later
macOS x86-64	12		Removed; use 2.9 LTS
Windows x86-64	10	**10**	Server 2019 and later
Kubernetes operator	1.26	**1.28**	Helm chart 5.x

2.9 LTS receives security fixes until 30 September 2027.

Upgrading from 2.x

1. Upgrade to 2.9.2 first and run your suite once; it warns about every deprecated flag.
2. Convert scenarios with `kestrel migrate scenarios/ --to toml --write`.
3. Install 3.0 and run `kestrel check` on every converted scenario.
4. Run once to create `kestrel.lock`, then commit it beside the scenarios.

Table 2. Flags and settings changed in 3.0

2.x	3.0	Notes
`--rps 500`	`--rate 500/s`	Units are required; per-minute rates use `/m`.
`--duration 10m`	`duration = "10m"`	Set in the scenario file; the flag still works until 4.0.
`scenario.yaml`	`scenario.toml`	Convert with `kestrel migrate`.
`--export statsd`	`export = "statsd"`	OpenTelemetry is the new default exporter.
`--graphite-host`		Removed with the Graphite exporter.
`--no-seed`		Removed: every run is seeded from `kestrel.lock`.

Questions are welcome in the discussion forum.

2.9.2 · 12 August 2026

- Warns when a scenario uses a flag removed in 3.0. (#2176)
- Fixed a crash when a target returned an empty `Content-Type` header. (#2171)

2.9.1 · 29 July 2026

- Security: updated the bundled TLS library to address CVE-2026-31337. (#2160)
- Histogram buckets above 60 s are no longer merged. (#2158)

Thanks

Kestrel 3.0 includes work from 64 contributors, 23 of them new. Special thanks to **Amara Okafor** for the arrival models, **Tomás Ribeiro** for the TOML migration, and **Yuki Tanaka** for the coordinated-omission fix. Kestrel is released under the MPL-2.0.